THE STACK OF COHERENT SHEAVES WITH PROPER SUPPORT IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1197. Standing situation for the coherent stack.
- Lemma lemma-coherent-diagonal (canonical tag 08W6). cited at proof line 1206. Representable diagonal of the coherent stack.
- Lemma algebraic-lemma-representable-diagonal (canonical tag 045G). cited at proof lines 1206-1207. Diagonal implies Isom sheaves.
- Lemma spaces-descent-lemma-fpqc-universal-effective-epimorphisms (canonical tag 04P2). cited at proof lines 1219-1220. Unique gluing of g_i.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof lines 1243-1244. Effectivity of the descent datum.
- Lemma spaces-morphisms-lemma-flat-permanence (canonical tag 05VY). cited at proof lines 1247-1248. Flatness of F over T.
- Lemma spaces-descent-lemma-finite-presentation-descends (canonical tag 060V). cited at proof lines 1249-1251. Finite presentation of F.
- Lemma spaces-descent-lemma-descending-property-proper (canonical tag 0422). cited at proof lines 1252-1253. Properness of the support descends.
- Lemma spaces-morphisms-lemma-flat-pullback-support (canonical tag 089C). cited at proof lines 1257-1258. Support commutes with flat base change.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10879–10879. Cited at statement line 1197: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 08W6: sources/stacks-chow/tags.txt, source lines 11259–11259. Cited at proof line 1206: Isom presheaves are sheaves via the diagonal; load-bearing.

Source reference. Original lemma 045G: sources/stacks-chow/tags.txt, source lines 5435–5435. Cited at proof lines 1206-1207: the transfer lemma; load-bearing.

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Source reference. Original lemma 04P2: sources/stacks-chow/tags.txt, source lines 6100–6100. Cited at proof lines 1219-1220: g_i glue uniquely to g : T -> B; load-bearing.

Source reference. Original proposition 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at proof lines 1243-1244: the descent datum (F_i, varphi_ij) is effective; load-bearing.

Source reference. Original lemma 05VY: sources/stacks-chow/tags.txt, source lines 7567–7567. Cited at proof lines 1247-1248: F is flat over T; load-bearing.

Source reference. Original lemma 060V: sources/stacks-chow/tags.txt, source lines 7739–7739. Cited at proof lines 1249-1251: F is of finite presentation; load-bearing.

Source reference. Original lemma 0422: sources/stacks-chow/tags.txt, source lines 5313–5313. Cited at proof lines 1252-1253: scheme-theoretic support of F proper over T; load-bearing.

Source reference. Original lemma 089C: sources/stacks-chow/tags.txt, source lines 10492–10492. Cited at proof lines 1257-1258: support commutes with the flat base change, feeding the properness descent; load-bearing.

Lemma 0.2. *In Situation 08KB the functor $p : \mathcal{C}oh_{X/B} \longrightarrow (Sch/S)_{fppf}$ is a stack in groupoids.*

Proof. To prove that $\mathcal{C}oh_{X/B}$ is a stack in groupoids, we have to show that the presheaves *Isom* are sheaves and that descent data are effective. The statement on *Isom* follows from Lemma 08W6, see Algebraic Stacks, Lemma 045G. Let us prove the statement on descent data. Suppose that $\{a_i : T_i \to T\}$ is an fppf covering of schemes over S. Let (ξ_i, φ_{ij}) be a descent datum for $\{T_i \to T\}$ with values in $\mathcal{C}oh_{X/B}$. For each i we can write $\xi_i = (T_i, g_i, \mathcal{F}_i)$. Denote $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ and $\mathrm{pr}_1 : T_i \times_T T_j \to T_j$ the projections. The condition that $\xi_i|_{T_i \times_T T_j} = \xi_j|_{T_i \times_T T_j}$ implies in particular that $g_i \circ \mathrm{pr}_0 = g_j \circ \mathrm{pr}_1$. Thus there exists a unique morphism $g : T \to B$ such that $g_i = g \circ a_i$, see Descent on Spaces, Lemma 04P2. Denote $X_T = T \times_{g,B} X$. Set $X_i = X_{T_i} = T_i \times_{g_i,B} X = T_i \times_{a_i,T} X_T$ and

$$X_{ij} = X_{T_i} \times_{X_T} X_{T_j} = X_i \times_{X_T} X_j$$

with projections pr_i and pr_j to X_i and X_j. Observe that the pullback of $(T_i, g_i, \mathcal{F}_i)$ by $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ is given by $(T_i \times_T T_j, g_i \circ \mathrm{pr}_0, \mathrm{pr}_i^* \mathcal{F}_i)$. Hence a descent datum for $\{T_i \to T\}$ in $\mathcal{C}oh_{X/B}$ is given by the objects $(T_i, g \circ a_i, \mathcal{F}_i)$ and for each pair i, j an isomorphism of $\mathcal{O}_{X_{ij}}$-modules

$$\varphi_{ij} : \mathrm{pr}_i^* \mathcal{F}_i \longrightarrow \mathrm{pr}_j^* \mathcal{F}_j$$

satisfying the cocycle condition over (the pullback of X to) $T_i \times_T T_j \times_T T_k$. Ok, and now we simply use that $\{X_i \to X_T\}$ is an fppf covering so that we can view $(\mathcal{F}_i, \varphi_{ij})$ as a descent datum for this covering. By Descent on Spaces, Proposition 04W8 this descent datum is effective and we obtain a quasi-coherent sheaf $\mathcal{F}$ over X_T restricting to $\mathcal{F}_i$ on X_i. By Morphisms of Spaces, Lemma 05VY we see that $\mathcal{F}$ is flat over T and Descent on Spaces, Lemma 060V guarantees that $\mathcal{F}$ is of finite presentation as an $\mathcal{O}_{X_T}$-module. Finally, by Descent on Spaces, Lemma 0422 we see that the scheme theoretic support of $\mathcal{F}$ is proper over T as we've assumed the scheme theoretic support of $\mathcal{F}_i$ is proper over T_i (note that taking scheme theoretic

support commutes with flat base change by Morphisms of Spaces, Lemma 089C). In this way we obtain our desired object over T. □